DATA ENGINEERING APPLICATIONS

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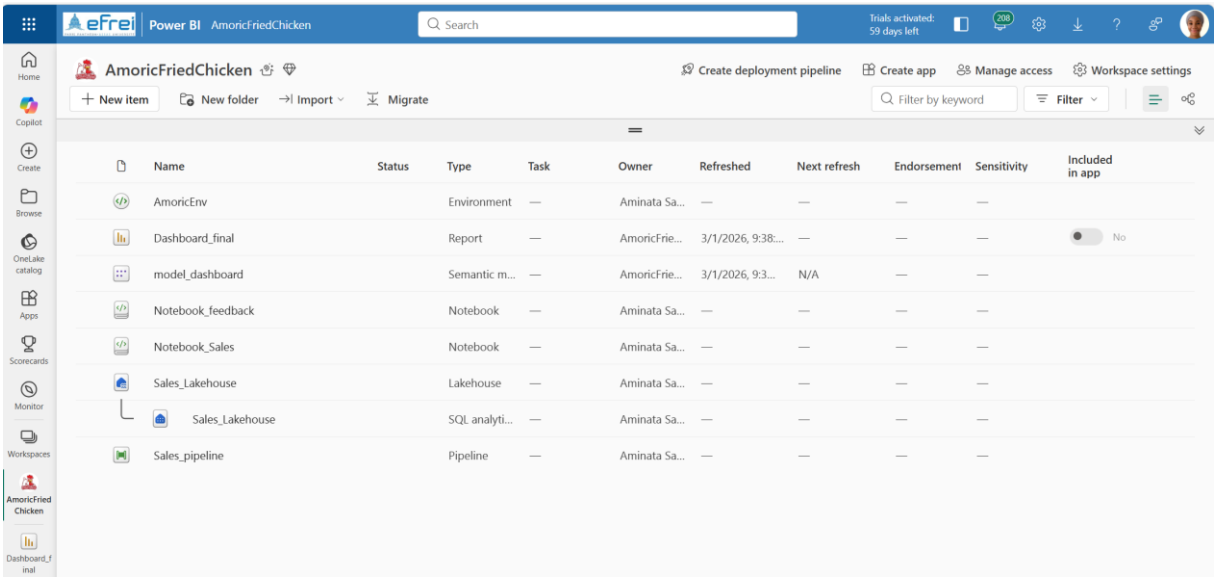
The aim of this project is to centralise, process and enhance two types of data that are essential for the Amoric Fried Chicken company : sales data and customer feedback. This information, initially available as CSV files deposited in a Teams space, must be transformed into a reliable, clean and easily usable dataset for analysis and decision-making.

To achieve this goal, we chose to build the entire compute in the Azure ecosystem, leveraging **Microsoft Fabric** for data ingestion, transformation, and storage, and **Power BI** for visualization and analysis. This choice was necessary for several reasons.

First, Microsoft Fabric offers a unified environment that brings together all the building blocks necessary for a data project: a centralized storage space (Lakehouse), automated ingestion pipelines, notebooks for processing, and Delta tables for high-performance and reliable storage. This integration avoids the multiplication of tools and simplifies project management. In addition, Fabric integrates naturally with Teams thanks to *Shortcuts*, which makes it possible to retrieve files uploaded by teams without duplication or complex manipulation.

Second, Power BI perfectly complements this architecture by allowing the Silver tables of the Lakehouse to be directly connected to create dynamic and interactive dashboards. The whole forms a coherent chain: the datais collected, cleaned, logged and then visualized in the same ecosystem, which guarantees technical continuity and great ease of use.

This report therefore presents the approach adopted to process Sales and Feedback data, the technical choices made, as well as the processing structure put in place to ensure an automated, reliable and scalable data flow.

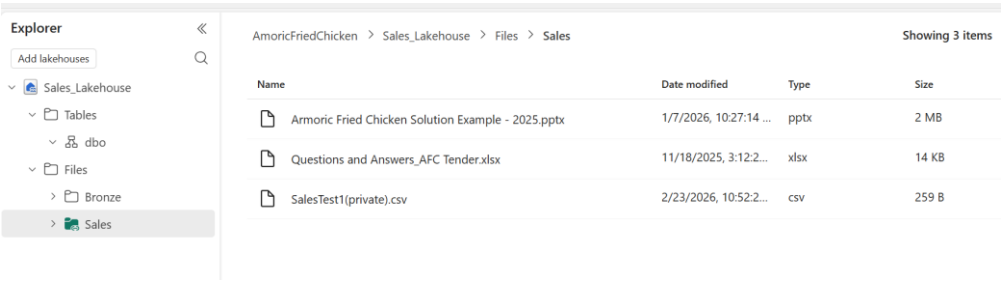


A- SALES DATA PROCESSING

The objective of the project is to set up an automated processing chain to ingest, clean, transform and log sales data from CSV files. The main challenge is to ensure reliable, automated ingestion and an actionable database for analysis and visualization.

A.1. Data source: Shortcut to Teams

CSV files are stored in a Teams folder. To use them in Microsoft Fabric, we create a **Shortcut Fabric.** This shortcut allows Fabric to "see" Teams files without copying them. This choice allows for real-time synchronization between the Teams folder and Fabric. So whenever the teams drop files into the folder, Fabric automatically retrieves them.

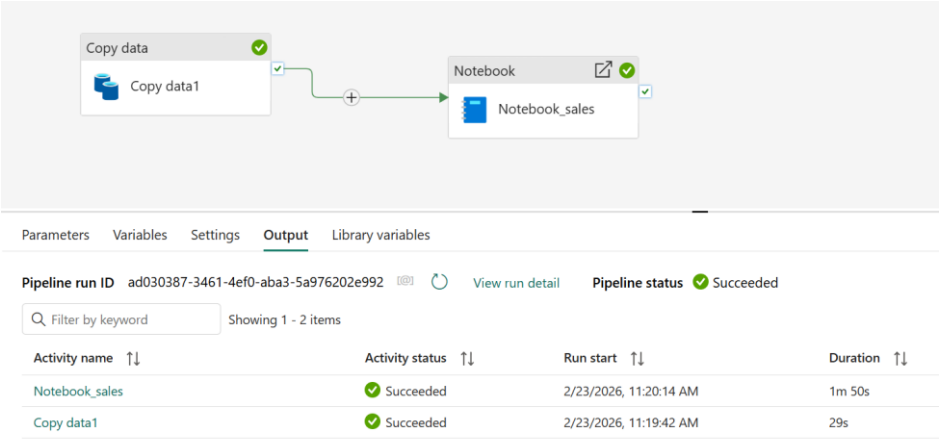


A.2. Organisation of processing

A pipeline is a small, automated program that runs on a schedule. Here, our pipeline has two main activities that clearly separate raw ingestion and processing:

- **Copy Data**: Copies CSV files from the Shortcut to the *Bronze layer* of the Lakehouse.
- **Notebook**: Runs a PySpark script to transform and clean up data to store in Delta tables in the *Silver layer*

This architecture follows modern data engineering standards, allowing raw data to be kept for audit or reprocessing and to produce clean, typed and deduplicated tables for Business Analysis.



A.3 Processing of data in the Notebook

- **Creation of the tracking table:** First a *file_tracking* table is created containing the information about the processed csv files. This table is essential to avoid reprocessing

the same files. It guarantees the idempotency of the pipeline, a fundamental principle in data engineering.

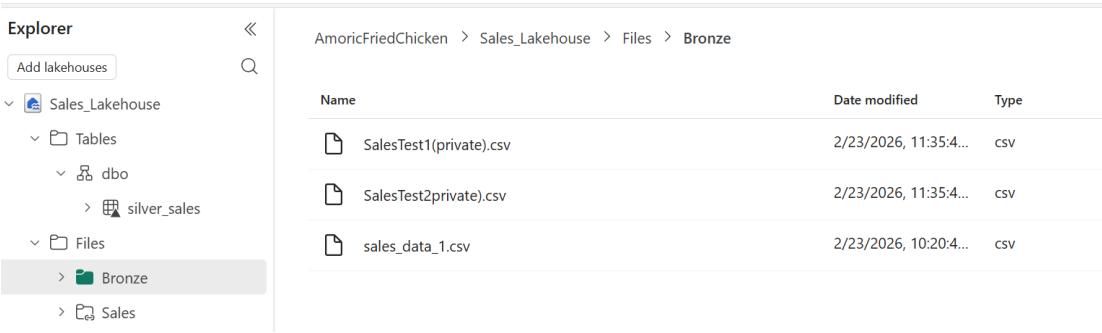
file_tracking

Showing 3 rows

Table view

	file_name	file_path	processed_at
1	SalesTest1(private).csv	abfss://3b672d63-a2ee-4726-92cb-a96df175d490@onelake.dfs....	2026-02-26T09:57:45.229Z
2	SalesTest2private).csv	abfss://3b672d63-a2ee-4726-92cb-a96df175d490@onelake.dfs....	2026-02-26T09:57:45.229Z
3	sales_data_1.csv	abfss://3b672d63-a2ee-4726-92cb-a96df175d490@onelake.dfs....	2026-02-26T09:57:45.229Z

- **Detecting new files:** Once the data has been inserted into the Bronze layer by the Copy Data activity, the notebook lists the files in the *Bronze* folder and then compares the paths with those already present in file_tracking in order to load only the new files and thus allow incremental processing.



- **Data loading and cleaning:** The csv files are then read with inferred schema and a cleanup is applied. For example, trim removes visible spurious spaces in the data (e.g. "Brazil|"), typing ensures consistency for calculations and BI, and removing duplicates avoids analysis errors.
- **Insert into Silver table:** The cleaned data is inserted into the Delta silver_sales table. The Delta format allows for ACID, time travel and optimal performance. The Silver table becomes the trusted source for dashboards.

silver_sales

Showing 1000 rows

Table view

	username	sale_date	country	product	quantity	unit_price	total_amount
1	user66	2025-07-25	Brazil	Grilled Tenders	13	4.71	61.23
2	user71	2025-09-26	Brazil	Chicken Nuggets	18	2.64	47.52
3	user309	2025-01-07	UK	Fried Wings	14	14.35	200.9
4	user375	2025-02-11	USA	Fried Wings	3	14.5	43.5
5	user63	2025-05-08	India	Spicy Strips	8	4.41	35.28
6	user468	2025-03-01	Brazil	Chicken Nuggets	3	12.1	36.3
7	user881	2025-02-26	India	Grilled Tenders	3	6.65	19.95
8	user873	2025-01-20	USA	Fried Wings	9	3.51	31.59
9	user956	2025-02-27	India	Spicy Strips	18	11.49	206.82
10	user692	2025-03-07	India	Grilled Tenders	1	4.57	4.57
11	user283	2025-07-28	Germany	Chicken Nuggets	19	3.3	62.7
12	user247	2025-01-05	USA	Chicken Nuggets	17	7.74	131.58
13	user176	2025-05-21	Brazil	Spicy Strips	19	2.61	49.59
14	user178	2025-08-20	France	Grilled Tenders	4	13.99	55.96
15	user4	2025-01-24	France	Spicy Strips	7	3.61	25.27

- **Updating the tracking table:** Each successfully processed file is added to the file_tracking table, which facilitates monitoring and allows a complete audit of the processing

At the end of the process:

- The sales files are correctly copied in Bronze.
- The tracking table ensures complete traceability
- The Silver table contains clean, typed data that is ready for analysis.
- The data is ready for use in Power BI or other analytics.

B- CAMPAIGN PROCESSING

B.1. Feedback Recovery

As part of marketing campaigns, it is essential to collect, centralize and analyze user feedback to:

- Measure customer satisfaction.
- Identify the strengths and areas for improvement of the campaigns.
- Feed an analytics pipeline to generate actionable insights.

The API was designed to meet these needs while ensuring simplicity, reliability, and scalability.

B.1.1 API URL

https://apiamoricfried-gvfzfwb6e2ewa6fr.swedencentral-01.azurewebsites.net/api/getfeedback

B.1.2 API Development and Components

Flask is used to structure the logic of the API. It allows you to manage HTTP routes (GET and POST) intuitively.

Pydantic for validation allows you to detect and reject non-compliant data and provide clear error messages to consumer systems.

REST architecture: The API exposes two operations:

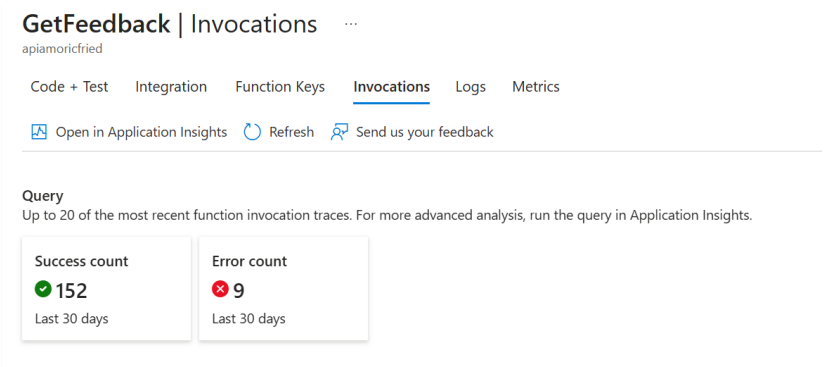
- **POST** : The API accepts either a single object or a list of feedback. The ability to send batches optimizes throughput during campaigns with many returns.
- **GET** - The API returns all the feedback collected

Azure Functions was used to deploy the API as a serverless function. This choice allows automatic server scalability. The platform instantly adapts to peak loads, especially during high-audience marketing campaigns, reducing manual maintenance and traditional server management. When a large number of POST requests arrive simultaneously, Azure Functions automatically creates new instances of the function, and each instance processes a portion of the requests. The ramp-up is almost instantaneous, without human intervention. Once the peak passes, unnecessary instances are removed to reduce costs.

Azure Blob Storage: Feedback is stored in an Azure Blob container, as a single JSON file, making it easy to read, export, and trace. The API adds each new feedback following the existing JSON file providing a complete historization. Blob Storage supports thousands of simultaneous writes, so even if a campaign generates tens of thousands of feedbacks, storage never becomes a saturation point.

A Planned Future Evolution is the ability to partition by date or by campaign.

In conclusion, the API has been designed to integrate seamlessly with the fabric pipeline. Feedback is collected as is, without transformation. The Bronze → Silver pipeline then takes care of the enrichment. This separation of roles guarantees maximum simplicity on the API side and controlled data quality.



B.2 Treatment and Enrichment

B.2.1. Purpose of the pipeline

The pipeline aims to collect, standardize, analyze, and store multilingual feedback from our API to produce an AI-enriched Silver table. In Microsoft Fabric, the pipeline follows a three-layer architecture (Bronze → Silver → enriched AI), inspired by the *Medallion Architecture* pattern. This structure makes it possible to clearly separate raw, normalized and enriched data and to make the pipeline robust, re-executable and industrializable.

B.2.2. Azure access security and configuration

Sensitive keys (OpenAI API, endpoint, feedback API URL) are stored in Azure Key Vault. This makes it possible to:

- Prevent accidental leakage into a repository
- Centralize secrets management
- Benefit from Azure RBAC access control, in this case *Key Vault Secrets User*

- To be able to revoke or renew secrets without modifying the code

Notebooks in Fabric thus dynamically retrieve secrets through the workspace's managed identity. The secrets currently used in the Key Vault are:

- AZURE-OPENAI-API-KEY
- AZURE-OPENAI-ENDPOINT
- FEEDBACK-URL

The notebook now uses *mssparkutils.credentials.getSecret()* to retrieve secrets. The use of Workspace Fabric managed identity also ensures passwordless authentication and compliance with Azure best practices

B.2.3. Bronze Ingestion: Raw Data and Deduplication

The Bronze table stores feedback exactly as the API returns it, without transformation. This ensures full traceability, avoids any loss of information, and replays the pipeline if necessary. However, each feedback receives a unique identifier based on a SHA256 hash built from: *username, campaign_id* and *Comment*

This choice is relevant because SHA256 is deterministic: same input → same output. Thus, it guarantees almost absolute uniqueness and makes it possible to detect duplicates even if the API returns the complete history.

Before inserting into Bronze, existing IDs are compared to new ones and only previously seen feedback is inserted. This mechanism thus avoids unnecessary reintegration and additional treatment costs.

B.2.4. Silver Standardization: Quality and Consistency

Normalization *normalize_date, normalize_campaign_id,* and *normalize_comment* features help harmonize dates, standardize campaign IDs, clean up comments, and eliminate inconsistent values. Invalid feedback is excluded from Silver, which ensures a reliable table and better quality for AI models.

B.2.5. AI analysis: robustness and enrichment

The goal of this step is to transform a raw marketing feedback comment into structured, enriched data. The tasks performed are:

- Translation of the commentary in English
- Overall score: From 0 to 5
- Sentiment analysis: "Positive, Negative, Neutral"
- Classification according to the marketing dimension: "Creativity, Clarity, Engagement, Global Strategy"
- Extraction of key adjectives: For example, "satisfied"

The Azure OpenAI GPT4.1 model is used to analyze multilingual feedback. This choice is justified by its ability to understand and translate several languages and its accuracy in sentiment analysis. This model was evaluated according to the consistency of the detected sentiments, the quality of the translations and the reproducibility of the results.

To check the reliability of the model, we conducted a **manual vs. AI test** on a sample of **70 feedbacks** from recent marketing campaigns. Each field in the model was compared with human annotation and precision was calculated as the percentage of exact matches.

Results:

Field	Accuracy
sentiment_label	97%
Marketing_dimension	98 %
key_adjectives	82 %

Observation:

- The model performs very well in terms of sentiment and marketing.
- Key adjectives have more variability, which is normal given the free nature of the words chosen

Conclusion

- Out of 70 feedbacks tested, the model showed a high overall accuracy, sufficient for operational use.
- The English translation was correct 100% of the time and facilitates multilingual analysis.
- The AI pipeline is validated to automatically enrich new feedback in the Silver table.

Processing flow:

For each new feedback received:

1. ***Retrieving raw information***
    - *feedback_id, username, feedback_date, campaign_id, comment.*
2. ***Creating the AI prompt***
    - *The feedback is sent to the model through Azure OpenAI.*
    - *The prompt explicitly requests:*
        - *Detect language.*
        - *Translate commentary to English*
        - *Perform sentiment analysis*
        - *Classify the marketing dimension*
        - *Extract a key adjective in English describing the feedback.*
    - *The expected result is strict **JSON** with these fields.*
3. ***Retrieving and parsing JSON***
    - *The JSON returned by the template is loaded in Python dictionary.*
    - *Fields are extracted: english_translation, marketing_dimension, star_rating (score calculated by AI), sentiment_label, key_adjectives*

4. ***Construction of the Delta Line***
    - *Each enriched feedback is stored in a dictionary with:*
        - *The original columns*
        - *The new AI-enriched Columns*
        - *Treatment time*

B.2.6. Idempotent Write via Delta MERGE

Before insertion into the Silver table, the AI results are filtered. We check that sentiment_label is in ["Positive", "Negative", "Neutral"], star_rating is between 0 and 5 and that the marketing_dimension is among the allowed values

The committed rows are then inserted into the Delta Silver table via a merge() on feedback_id. This choice guarantees that only new feedback is inserted and that it is unique

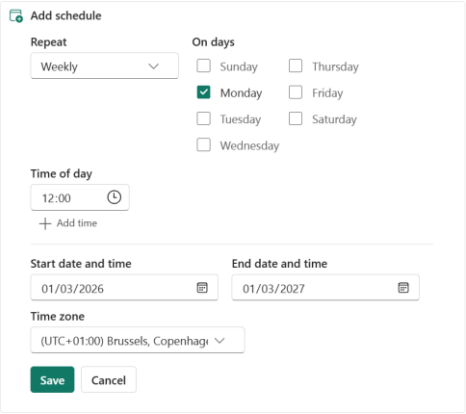
Each row in silver_feedback now contains:

Column	Description
feedback_id	Unique feedback identifier
username	Username
feedback_date	Feedback date
campaign_id	Marketing Campaign ID
How	Original text
english_translation	Commentary translated into English
marketing_dimension	Marketing dimension analysed
star_rating	Score 0-5 given by the AI
sentiment_label	Sentiment detected
key_adjectives	Main keyword describing the feedback
processed_at	Date and time of treatment

B.2.7. Automation

Feedback can arrive at any time, in small or very high quantities, depending on the marketing activity. The API deployed on Azure Functions is designed to absorb these variations without human intervention, because Azure Functions automatically creates as many instances as needed as the volume increases.

Unlike collection, AI processing does not need to be continuous. It is deliberately decoupled and planned, which provides stability and cost control. Indeed, making an AI call for each individual comment would increase expenses. This is why during periods of high marketing activity, the customer can easily adjust the frequency of execution of the AI processing. The Fabric scheduler allows you to switch from weekly to daily processing, or even multiple times a day.



This combination of continuous collection and planned treatment is a strategic choice that brings several benefits:

- **Resiliency** : The API absorbs spikes without the risk of saturation.
- Cost containment: AI is only executed when necessary.
- **Flexibility** : the frequency can be adjusted according to marketing needs.
- **Operational simplicity** : teams don't have to manage scalability.
- **Data quality** : Batch processing ensures consistent normalization.

C- BUSINESS ANALYTICS

C.1. Semantic Model

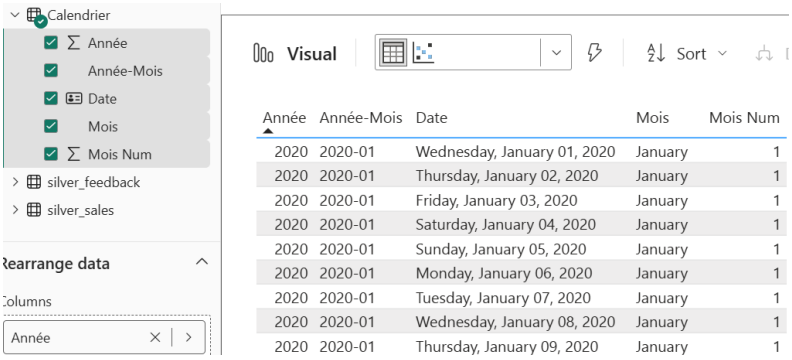
The model is based on three main tables.

Sales table*:* This table contains transactional data related to sales. It is a **table of facts**, with potentially high volume and quantitative measures.

Feedback *table*: This table gathers user feedback on marketing campaigns. It is also a **table of facts**, focusing on the perceived quality of campaigns.

Calendar table: A date table has been created to structure the temporal analyses. It contains: *Date, Year* and *Month*. This table constitutes a temporal dimension, essential for chronological analyses. The creation of this date table makes it possible to:

- Standardize temporal analyses between sales and feedback,
- Facilitate monthly, quarterly and annual comparisons,
- Use time-based DAX functions (YTD, YOY, MOM),
- Ensure the consistency of temporal filters in visuals.



Année	Année-Mois	Date	Mois	Mois Num
2020	2020-01	Wednesday, January 01, 2020	January	1
2020	2020-01	Thursday, January 02, 2020	January	1
2020	2020-01	Friday, January 03, 2020	January	1
2020	2020-01	Saturday, January 04, 2020	January	1
2020	2020-01	Sunday, January 05, 2020	January	1
2020	2020-01	Monday, January 06, 2020	January	1
2020	2020-01	Tuesday, January 07, 2020	January	1
2020	2020-01	Wednesday, January 08, 2020	January	1
2020	2020-01	Thursday, January 09, 2020	January	1

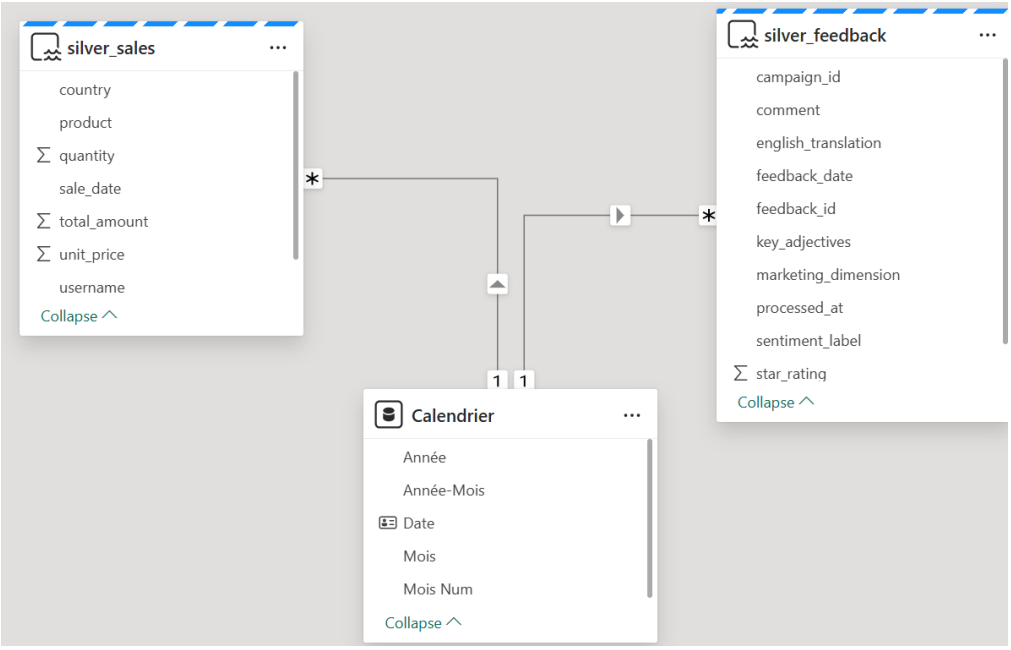
Thus, two main relationships have been created between the Calendar table and the fact tables:

3.1. Calendar relationship→ silver_sales

- *Linking column: Calendar[Date] → silver_sales[sale_date]*
- *Cardinality: Many-to-one (:1)**

3.2. Calendar Relationship→ silver_feedback

- *Linking column: Calendar[Date] → silver_feedback[feedback_date]*
- *Cardinality: Many-to-one (:1)**



These relationships follow the best practices of a **star model**, where the Calendar table acts as the central dimension. The *silver_sales* and *silver_feedback* tables contain multiple rows per date, while the Calendar table contains a single row per date. The *:1 cardinality is therefore the only correct one and guarantees reliable analytical behavior.

C.2. Dashboard

The dashboard has been designed to provide an integrated view of sales performance, customer satisfaction, and marketing campaign effectiveness. It combines financial indicators, time

analytics, qualitative insights, and geographic breakdowns, enabling informed and timely decision-making.

Financial and volumetric indicators

- **Total Revenue:** This metric answers the question: *"What is the total revenue generated over the selected period?"* It allows you to immediately assess the overall financial performance.
- **Sum of Quantity** This KPI answers " *What total volume of products was sold?"* It offers a measure of the level of activity independent of prices.

Analysis by product

- **Sum of total_amount by Product** This graph answers *"Which products generate the most revenue?" "Which products perform better or worse?"* It helps identify bestsellers and products that need attention.

Geographic analysis

- **Sum of total_amount by country** This visual answers the question of: *"Which countries contribute the most to sales?" "What is the geographical distribution of sales?"* It highlights strategic markets and areas with potential.

Customer Sentiment Analysis

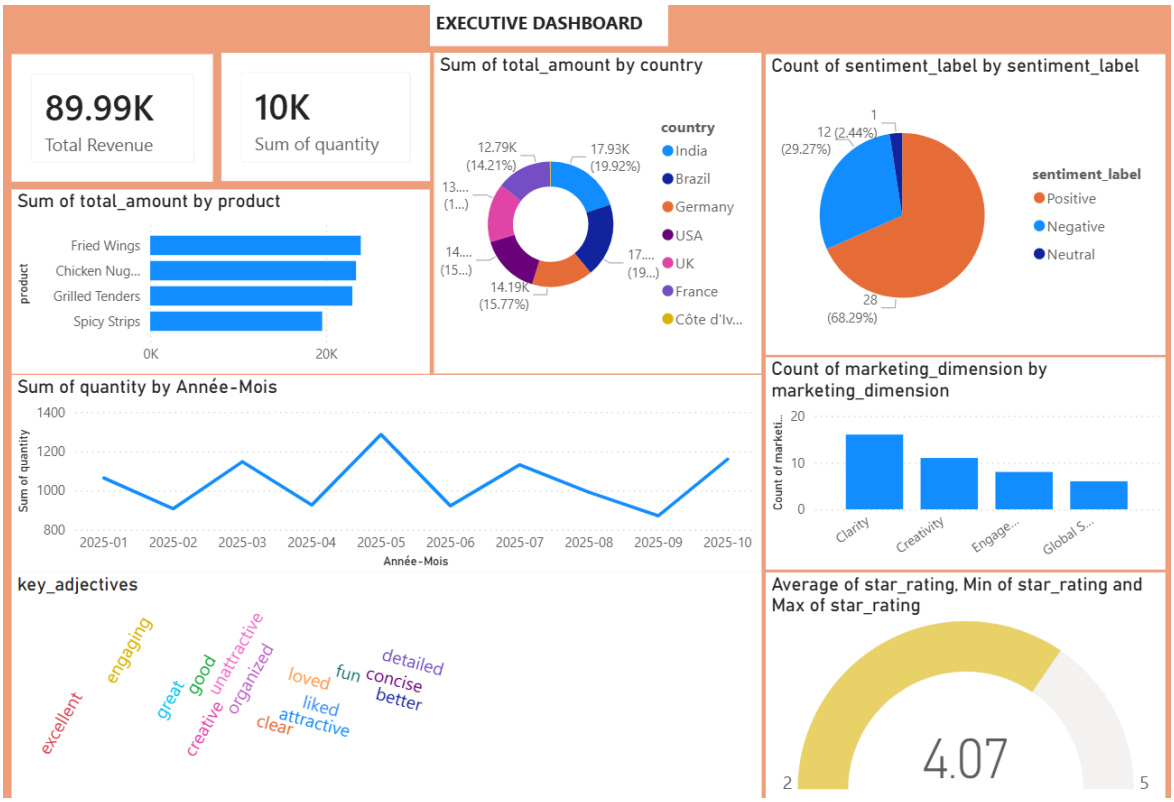
- **Count of sentiment_label by sentiment_label** This graph answers the *question: "What is the overall sentiment expressed by customers?" "What proportion of feedback is positive, negative or neutral?"* It helps to assess the overall perception of campaigns and products.
- **Average / Min / Max star_rating** This metric answers to: *"What is the average rating given by customers?" "What is the best and worst grade observed?"* It provides a synthetic measure of customer satisfaction.
- **Word Cloud – Key Adjectives** This word cloud answers *"Which adjectives come up most often in customer reviews?"* It sheds light on dominant themes and key perceptions.

Temporal analysis

- **Sum of quantity by Year-Month** This graph answers to: *"How do sales evolve over the months?" "Are we observing seasonality, growth or decline?"* It allows you to monitor the commercial dynamics over time.

Marketing Dimensions Analysis

- **Count of marketing_dimension by marketing_dimension** This visual answers the question: *"Which marketing aspects are most often mentioned in feedback?"* It helps identify the most visible or sensitive dimensions for customers.



C.3. Role of filters in analysis

3.1. Year filter

This filter allows you to respond to:

- *"How do sales and feedback evolve from year to year?"*
- *"Which year was the most successful?"*
- *"Is customer sentiment improving or deteriorating over the years?"*

It synchronizes the entire dashboard to offer consistent temporal analysis.

3.2. Campaign ID filter

This filter allows you to respond to:

- *"Which marketing campaign generated the most sales?"*
- *"Which campaign received the best or worst feedback?"*
- *"How does performance vary from campaign to campaign?"*

It allows for granular analysis, focused on the effectiveness of marketing actions.

Conclusion

This project has made it possible to set up a complete and reliable processing chain to centralize, clean and enhance sales (*Sales*) and customer feedback data. Leveraging the Azure ecosystem, and more specifically Microsoft Fabric and Power BI, we built a solution that was consistent, modern, and perfectly suited to the analytics needs of the business.

Using Microsoft Fabric provided a unified environment to manage data ingestion, storage, and transformation. The Bronze and Silver layered structuring made it possible to clearly separate the raw data from the data ready for analysis, while ensuring better quality and traceability. The implementation of a monitoring table for processed files has increased the reliability of the process by avoiding duplicates and ensuring a complete history of processing.

Power BI was a natural fit into this architecture by enabling Silver data to be transformed into dynamic and accessible dashboards. This continuity between Fabric and Power BI has facilitated the creation of relevant indicators, useful for sales management and the analysis of customer feedback.

Ultimately, this project demonstrates that a well-thought-out architecture, based on integrated and automated tools, can transform dispersed CSV files into a true source of strategic information. The solution implemented is scalable, robust and ready to accommodate new needs, whether it is more advanced analytics, predictive models or new data sources.